Sensory Afferent Database: Advancing towards a comprehensive understanding of sensory afferents and motor control

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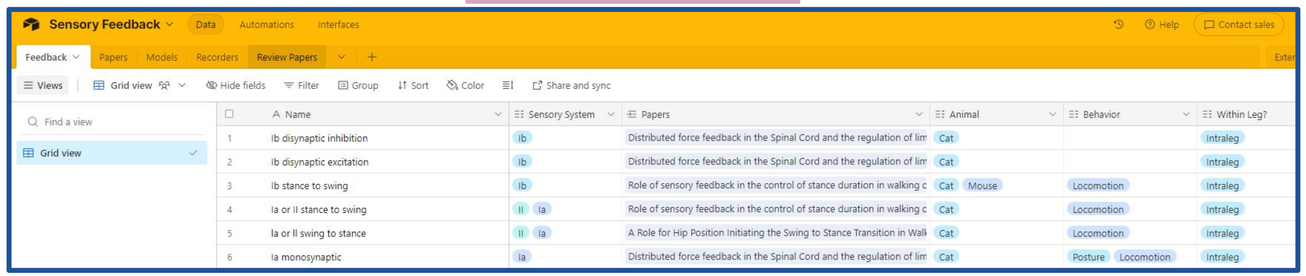
Abstract

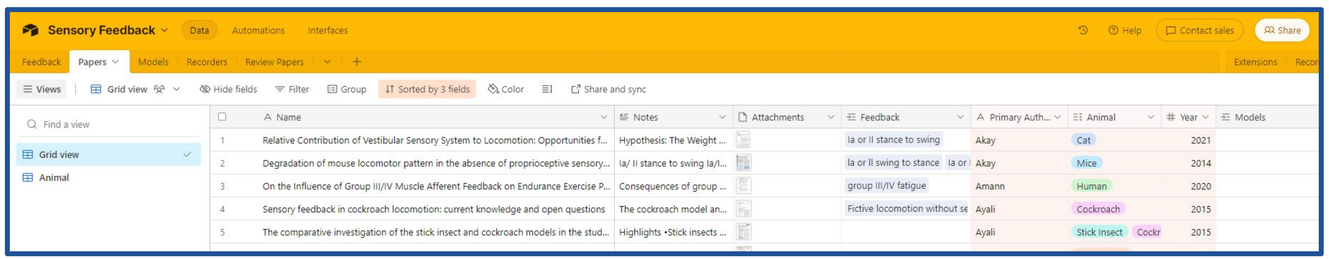
The Sensory Afferent Database is a pioneering tool aimed at facilitating comprehensive research on the interplay between sensory afferents and motor control. This large-scale database, currently hosted on Airtable, offers researchers a centralized platform to access and share papers that investigate the influence of afferent feedback (e.g., mechanosensory and proprioceptive) on locomotion in diverse animal species. Its primary objective is to include all high-quality papers that discuss the intricate relationship between sensory afferent feedback and motion control.

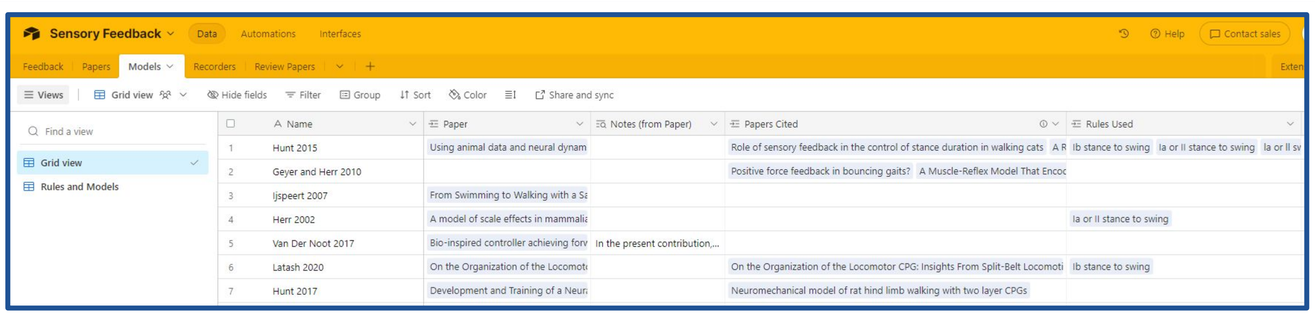
The database's user-friendly interface allows for seamless navigation and exploration. Researchers can organize papers based on categories, including models, review papers, and animal studies. Furthermore, the papers are tagged with essential metadata, such as sensory system involved (e.g. Ib, Chordontonal Organs), animal species used or modeled (e.g. cat, cockroach), feedback rules modeled (e.g. Ib stance to swing transition, Ia inhibition), behavior studied (e.g., posture or locomotion), and whether the feedback effect is intraleg or interleg.

With an extensive collection of papers encompassing various research categories, the Sensory Afferent Database facilitates multidisciplinary collaboration and encourages the exchange of ideas among researchers investigating organism motion control. This database may be used as an instructional tool to train both new and experienced scientists as to the different sensory systems used in a wide variety of animals. Currently, this tool is primarily used within the NeuroNex group Communication, Coordination, and Control in Neuromechanical Systems to compare and contrast feedback mechanisms used by animals in different phyla. By harnessing this diverse range of literature, we are working to gain insights into the governing mechanisms that facilitate effective neural-motor control.

Screenshots of the database:







Next Steps

We will continue expanding the database by adding papers, models, review papers, and feedback types. The current entries will also be more fully filled out.

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